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B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, JULY 2024
Third Semester

21CSC202J – OPERATING SYSTEM

(For the candidates admitted from the academic year 2022-2023 onwards)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- (ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

PART – A (20 × 1 = 20Marks)

Marks BL CO PO

Answer **ALL** Questions

- | | | | | |
|--|---|---|---|---|
| 1. To access the services of the operating system, the interface is provided by the _____. | 1 | 2 | 1 | 1 |
| (A) Library | | | | |
| (B) System calls | | | | |
| (C) Assembly instructions | | | | |
| (D) API | | | | |
| 2. _____ of the following is not true. | 1 | 2 | 1 | 1 |
| (A) Kernel is the program that constitutes the central core of the operating system | | | | |
| (B) Kernel is the first part of operating system to load into memory during booting | | | | |
| (C) Kernel is made of various modules with cannot be loaded in running operating system | | | | |
| (D) Kernel remains in the memory during the entire computer session | | | | |
| 3. A _____ is an active entity whereas _____ is said to be a passive entity. | 1 | 1 | 1 | 1 |
| (A) Process, program | | | | |
| (B) Program, process | | | | |
| (C) Processor, process | | | | |
| (D) Program, processor | | | | |
| 4. The time taken to switch between user and kernel modes of execution be t_2 while the time taken to switch between two processes be t_1 . Identify the true statement. | 1 | 2 | 1 | 1 |
| (A) $t_2 > t_1$ | | | | |
| (B) $t_2 = t_1$ | | | | |
| (C) $t_2 < t_1$ | | | | |
| (D) Nothing can be said about the relation between t_1 and t_2 | | | | |
| 5. List of processes that are prepared to be executed and waiting are in _____. | 1 | 1 | 2 | 1 |
| (A) Job queue | | | | |
| (B) Ready queue | | | | |
| (C) Execution queue | | | | |
| (D) Process queue | | | | |
| 6. The segment of code in which the process may change common variables, update tables, write into file is known as _____. | 1 | 2 | 2 | 1 |
| (A) Program | | | | |
| (B) Non-critical section | | | | |
| (C) Critical section | | | | |
| (D) Synchronizing | | | | |

7. _____ is referred to as the collection of data section, text section stack and memory limits of a process. 1 1 2 1
 (A) Process control block (B) Process state block
 (C) Process status block (D) Process information block
8. _____ of the following is called job scheduling? 1 1 3 1
 (A) Long-term scheduling (B) Short-term scheduling
 (C) CPU scheduling (D) Medium term scheduling
9. Consider several processes access the same data concurrently and the outcome of the execution depends on the particular order in which the access takes place is called _____. 1 2 3 1
 (A) Dynamic condition (B) Race condition
 (C) Essential condition (D) Critical condition
10. A semaphore is a shared integer variable and _____. 1 1 3 1
 (A) That can not drop below zero (B) That can not be more than zero
 (C) That can not drop below one (D) That can not be more than one
11. Round Robin scheduling is _____ in a time shared system. 1 2 3 1
 (A) Using very large time slices converts it into first come first served scheduling algorithm
 (B) Using very small time slices converts it into first come first served scheduling algorithm
 (C) Using extremely small time slices increases performance
 (D) Using very small time slices converts into shortest job first algorithm
12. Mutual exclusion implies that _____. 1 2 3 1
 (A) If a process is executing in its critical section, then no other process must be executing in their critical section
 (B) If a process is executing in its critical section, then other processes must be executing in their critical sections
 (C) If a process is executing in its critical section, then all the resources of the system must be blocked until it finishes execution
 (D) Process not executing in critical section
13. In the bounded buffer problem _____. 1 1 4 1
 (A) There is only one buffer (B) There are 'n' buffers (n being greater than one but finite)
 (C) There are infinite buffers (D) The buffer size is bounded
14. The Dining-Philosophers problem will occur in case of _____. 1 1 4 1
 (A) 5 philosophers and 5 chopsticks (B) 4 philosophers and 5 chopsticks
 (C) 3 philosophers and 5 chopsticks (D) 6 philosophers and 5 chopsticks
15. The situation where the processor spends most of its time in swapping process pieces rather than execution instruction is called _____. 1 2 4 1
 (A) Paging (B) The principle of locality
 (C) Thrashing (D) Swapping
16. _____ of the following is the main drawback of FIFO page replacement algorithm? 1 1 4 1
 (A) Requirement of large memory (B) Frame allocation
 (C) Reduction in multiprogramming (D) Reduced optimality

17. The Light Directory Access Protocol (LDAP) doesn't store _____.
 (A) Users (B) Address
 (C) Passwords (D) Security keys
18. _____ is an approach to restrict system access to authorized users.
 (A) Role-based access control (B) Process-based access control
 (C) Job-based access control (D) Program based access control
19. The bankers algorithm is _____ than the resource allocation graph algorithm.
 (A) Less efficient (B) More efficient
 (C) Equal (D) Zero
20. A monitor is a type of _____.
 (A) Semaphore (B) Low level synchronization construct
 (C) High level synchronization construct (D) Medium level synchronization construct

PART – B (5 × 8 = 40 Marks)

Answer **ALL** Questions

Marks BL CO PO

21. a. Explain the essential properties of the following operating systems.
 (i) Batch 2
 (ii) Time sharing 2
 (iii) Parallel 2
 (iv) Distributed 2

(OR)

- b.i. Explain in detail about the various reasons involved in process creation and termination. 4 3 1 1
- ii. Compare mode switching and process switching. 4 3 1 1
22. a. Describe the solution using semaphores to solve dining philosopher problem. 8 4 2 3

(OR)

- b. Discuss how wait () and signal () semaphore operations should be implemented in multiprocessor environments, using test and set instructions. The solution should exhibit minimal busy waiting. Develop pseudo code for implementing operations. 8 4 2 3
23. a. Consider the following set of processes with the length of the CPU-burst time in given ms. 8 4 3 3

Process	Arrival time	Burst time
P1	8	0
P2	4	1
P3	9	2
P4	5	3
P5	3	4

Draw four Gantt charts illustrating the execution of these processes using FCFS, SJF, priority scheduling. Also calculate waiting time and turn around time for each scheduling algorithms.

(OR)

- b. Compare and explain internal fragmentation and external fragmentation. 8 4 3 3
24. a. Given six memory partitions of 300 kB, 600 kB, 350 kB, 200 kB, 750 kB and 125 kB (in order). How would the first-fit, best-fit and worst-fit algorithms place processes of size 115 kB, 500 kB, 358 kB, 200 kB and 375 kB (in order)? Rank the algorithms in terms of how efficiently they use memory. 8 4 4 3

(OR)

- b. Consider page reference string 1, 3, 0, 3, 5, 6, 3 with 3 page frames. Find the number of page faults. 8 4 4 3
25. a. Describe the following in detail with respect to disk: 8 3 5 2
- (i) Seek time
 - (ii) Latency
 - (iii) Access time
 - (iv) Transfer time

(OR)

- b. Discuss various file organization methods with an example. 8 3 5 2

PART – C (1 × 15 = 15 Marks)

Answer **ANY ONE** Question

Marks BL CO PO

26. Consider a system that contains five processes P1, P2, P3, P4, P5 and three resources types A, B and C. The resources types A has 10, B has 5 and C has 7 instances. Answer the following questions using the banker's algorithm. 15 4 4 3

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P1	0	1	0	7	5	3	3	3	2
P2	2	0	0	3	2	2			
P3	3	0	2	9	0	2			
P4	2	1	1	2	2	2			
P5	0	0	2	4	3	3			

- (i) Construct the need matrix.
 - (ii) Determine if the system is safe or not.
27. Consider that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive is currently serving request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order is 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130. Starting from the current head position, find is the total distance (in cylinders) that the disk arm moves to satisfy all pending, request for each of the following disk scheduling algorithm: 15 4 5 3
- (i) FCFS
 - (ii) SSTF
 - (iii) SCAN
 - (iv) C-SCAN
 - (v) LOOK
 - (vi) C-LOOK

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